Specical topics in EM Algorithm

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Contents

1 Simple Version

1.1 Motivating example

1.1.1 Setup

Given observed data $X_1, ..., X_n$, assume they are independently and identically distributed (iid) from some mixture distribution:

$$X_i, ..., X_n \overset{\text{iid}}{\sim} F(X|\Theta)$$

This distribution is a Gaussian Mixture Model (GMM):

$$f(X|\Theta) = \sum_{k=1}^{K} \pi_k N(X|\mu_k, \sigma_k^2)$$

Where:

- π_k are the mixture weights satisfying $\sum_{k=1}^{K} \pi_k = 1$.
- Θ is the parameter set, including:

$$\Theta = \{\pi_k, \mu_k, \sigma_k^2\}_{k=1}^{K}$$

1.1.2 Likelihood Function

$$L(\Theta) = \prod_{i=1}^{n} \sum_{k=1}^{K} \pi_k N(X_i|\mu_k, \sigma_k^2)$$

Taking the logarithm:

$$\log L(\Theta) = \sum_{i=1}^{n} \log \left(\sum_{k=1}^{K} \pi_k N(X_i|\mu_k, \sigma_k^2) \right)$$

Since the logarithm operates on a summation, differentiation becomes challenging, making direct optimization difficult.

1.2 Alternative Approach

1.2.1 Introducing Latent Variables

Define latent variables Z_i, indicating which component each data point belongs to:

$$P(Z_i = k) = \pi_k$$

Given $Z_i = k$, the data X_i follows:

$$X_i | Z_i = k \sim N(\mu_k, \sigma_k^2)$$

Thus, the joint likelihood can be rewritten as:

$$L(\Theta) = \prod_{i=1}^{n} f(X_i|Z_i)P(Z_i)$$

Since Z_i are independent:

$$L(\Theta) = \prod_{i=1}^{n} P(Z_i) \prod_{i=1}^{n} f(X_i|Z_i)$$

That is:

$$L(\Theta) \propto \prod_{i=1}^{n} N(X_i|\mu_{Z_i}, \sigma_{Z_i}^2)$$

Taking the logarithm:

$$\log L(\Theta) = \sum_{i=1}^{n} \log N(X_i|\mu_{Z_i}, \sigma_{Z_i}^2)$$

1.2.2 Expanding the Log-Likelihood

The log density of a normal distribution:

$$\log N(X_i|\mu, \sigma^2) = -\frac{1}{2}\log(2\pi\sigma^2) - \frac{(X_i - \mu)^2}{2\sigma^2}$$

Thus:

$$\log L(\Theta) = \sum_{i=1}^{n} \left[-\frac{1}{2}\log(2\pi\sigma_{Z_i}^2) - \frac{(X_i - \mu_{Z_i})^2}{2\sigma_{Z_i}^2} \right]$$

Rearranging the summation:

$$= \sum_{k=1}^{K} \sum_{i:Z_i=k} \left[-\frac{1}{2}\log(2\pi\sigma_k^2) - \frac{(X_i - \mu_k)^2}{2\sigma_k^2} \right]$$

Remark: The key idea is to **group terms by mixture component** k, rather than summing directly over all data points i.

Original Log-Likelihood The log-likelihood function is initially written as:

$$\log L(\Theta) = \sum_{i=1}^{n} \left[-\frac{1}{2}\log(2\pi\sigma_{Z_i}^2) - \frac{(X_i - \mu_{Z_i})^2}{2\sigma_{Z_i}^2} \right]$$

where Z_i denotes the component assignment of X_i.

Rearranging the Summation To simplify computation, we reorganize the summation by first summing over mixture components k, then summing over data points assigned to each component:

$$\sum_{k=1}^{K} \sum_{i:Z_i=k} \left[-\frac{1}{2} \log(2\pi\sigma_k^2) - \frac{(X_i - \mu_k)^2}{2\sigma_k^2} \right]$$

Why this Works?

- Instead of summing over all i first, we group terms by component k.
- This transformation separates the contribution of each component, making parameter estimation clearer.
- It aligns with the Expectation-Maximization (EM) algorithm's M-step, where parameters are updated component-wise.

1.2.3 Parameter Derivation (Maximization Step)

Estimating μ_k Taking the derivative with respect to μ_k and setting it to zero:

$$\frac{\partial L(\Theta)}{\partial \mu_k} = 0$$

Solving for μ_k:

$$\hat{\mu}_k = \frac{1}{n_k} \sum_{i:Z_i=k} X_i$$

where n_k is the number of samples assigned to the k-th component.

Estimating σ_k^2 Taking the derivative with respect to σ_k^2 and setting it to zero:

$$\frac{\partial L(\Theta)}{\partial \sigma_k^2} = 0$$

Solving for σ_k^2:

$$\hat{\sigma}_k^2 = \frac{1}{n_k} \sum_{i:Z_i=k} (X_i - \hat{\mu}_k)^2$$

1.2.4 Conclusion

This section primarily illustrates:

- Directly optimizing the log-likelihood of a Gaussian Mixture Model is difficult because the logarithm applies to a summation.
- By introducing latent variables Z_i, the problem is reformulated into a more manageable form, making parameter estimation more intuitive.

- The final parameter estimates match the M-step results in the Expectation-Maximization (EM) algorithm.

This is the core idea behind the EM algorithm: using latent variables Z_i to transform a difficult log-likelihood function into an easier optimization problem.

1.3 EM Algorithm workflow

1.3.1 Definition

- Let the **complete data** be $\{X, Z\}$, where X is the observed data and Z is the hidden variable.
- The **complete data likelihood** function is:

$$f(X, Z|\Theta)$$

- The **incomplete data likelihood** function is:

$$f(X|\Theta)$$

  where X is incomplete because the hidden variable Z is unobserved.

1.3.2 Two Cases

- **If Z is known**, we can directly compute the MLE from the **complete data likelihood** $f(X, Z|\Theta)$.
- **If Z is unknown**, we need to use the **EM algorithm**.

1.3.3 Main Steps of EM Algorithm

Expectation Step (E-step):

$$\mathbb{E}_{Z|X,\Theta^{(t)}} [\log f(X, Z|\Theta)] = Q(\Theta, \Theta^{(t)}) \tag{1}$$

where:

- The expectation is taken over the hidden variable Z, given the current parameter estimate $\Theta^{(t)}$.
- The goal is to construct the auxiliary function $Q(\Theta, \Theta^{(t)})$.

1.3.4 Iteration and Updates

At the t-th iteration, we have the current estimate:

$$\Theta^{(t)}$$

E-step: Compute the **posterior distribution** of the hidden variable:

$$Z|X, \Theta^{(t)}$$

And use it to compute the function:

$$Q(\Theta, \Theta^{(t)})$$

M-step: Update parameters by maximizing $Q(\Theta, \Theta^{(t)})$:

$$\Theta^{(t+1)} = \arg\max_{\Theta} Q(\Theta, \Theta^{(t)})$$

1.3.5 Convergence Criterion

- Continue iterating until the parameter estimates $\Theta^{(t)}$ converge.
- The algorithm terminates when the parameter updates become sufficiently small, indicating the likelihood has stabilized.

1.3.6 Summary

- EM alternates between the **E-step** (computing the expectation of the complete log-likelihood) and the **M-step** (maximizing it).
- This process iteratively refines parameter estimates and converges to the Maximum Likelihood Estimate (MLE).
- In Gaussian Mixture Models (GMMs), Z_i represents the probability of each data point belonging to a component, and EM exploits this latent information for optimization. The different latent variables Z_i and parameters Θ for Normal distributions represent different mixture distributions. We can find the best GMM to fit the data (having the largest likelihood) through the EM algorithm.

1.4 Expectation-Maximization for Gaussian Mixture Models (Returning to example)

1.4.1 Problem Setup

We revisit the Gaussian Mixture Model (GMM) example:

$$X_i | Z_i = k \sim N(\mu_k, \sigma_k^2)$$

where Z_i is the latent variable indicating which Gaussian component X_i belongs to.

The **complete data likelihood** is:

$$f(X, Z|\Theta) = \prod_{i=1}^{n} \prod_{k=1}^{K} \left[\pi_k N(X_i|\mu_k, \sigma_k^2) \right]^{Z_{i,k}}$$

where $Z_{i,k}$ is an indicator variable.

Taking the logarithm:

$$\log f(X, Z|\Theta) = \sum_{i=1}^{n} \sum_{k=1}^{K} Z_{i,k} \left[\log \pi_k + \log N(X_i|\mu_k, \sigma_k^2)\right]$$

Where:

$$N(X_i|\mu_k, \sigma_k^2) = \frac{1}{\sqrt{2\pi\sigma_k^2}} \exp\left(-\frac{(X_i - \mu_k)^2}{2\sigma_k^2}\right)$$

1.4.2 E-Step

Posterior Probabilities Update The posterior probability of $Z_i = k$ given X_i is:

$$\begin{aligned} P(Z_i = k|X_i, \Theta^{(t)}) &= \frac{f(X_i|Z_i = k, \Theta^{(t)})\pi(Z_i = k|\Theta^{(t)})}{f(X_i|\Theta^{(t)})} \\ &= \frac{f(X_i|Z_i = k, \Theta^{(t)})\pi(Z_i = k|\Theta^{(t)})}{\sum_{l=1}^{K} f(X_i|Z_i = l, \Theta^{(t)})\pi(Z_i = l|\Theta^{(t)})} \\ &= \frac{f(X_i|Z_i = k, \Theta^{(t)})\pi_k^{(t)}}{\sum_{l=1}^{K} f(X_i|Z_i = l, \Theta^{(t)})\pi_l^{(t)}} \end{aligned}$$

Remark-1: Recall the notation for random variable in frequentists, X can also be rewritten as $X|\Theta$, where Θ is known. Here, under t+1 times iteration for E-step, we have the known GMM distribution parameters Θ^t. So, we should use the Bayesian point of view to look at $\pi(Z_i = k|\Theta^{(t)})$.

Rmark-2: Using Bayes theorem, Z_i is a discrete random variable. The prior probability for $Z_i = k$ is $\pi(Z_i = k|\Theta^{(t)})$. So, we can rewrite $f(X_i|\Theta^{(t)})$ as $\sum_{l=1}^{k} f(X_i, Z_l|\Theta^{(t)})$ (Joint distribution to Marginal distribution by definition). Further, we have $\sum_{l=1}^{k} f(X_i, Z_l|\Theta^{(t)}) = \sum_{l=1}^{K} f(X_i|Z_i = l, \Theta^{(t)})\pi(Z_i = l|\Theta^{(t)})$.

Remark-3: Like what we have discussed in the notation for frequentists, you can also rewrite the prior $\pi(Z_i = k|\Theta^{(t)})$ as $\pi(Z_i = k)$. But you should keep in mind, here $\pi(Z_i = k)$ is equal to π_k^t rather than the initial setting π_k.

Remark-4: This is the beauty of Bayesian Statistics. When you have the new information, you can update the prior. Then, you can regard the latest posterior as your prior, and use the new information to update it. This sequential updating process works in the EM algorithm because we can continuously have the new information for parameters Θ owing to the M step until convergence. Although, the information for X remains fixed. The new information for each cycle comes from the M step for parameters Θ.

Remark-5: The last thing I want to mention is the difference between data and information. The word information has a broader range than the word data. In the GMM E-step, information comes from the data X and parameters Θ together. But in the Bayes setting, the posterior comes from the conditional probability based on data. Since you need the data to get the likelihood part. Then based on this clarification, it's clearly that $\pi(Z_i = k|\Theta^{(t)})$ is not a posterior probability with respect to data X_i.

Substituting the Gaussian density:

$$P(Z_i = k|X_i, \Theta^{(t)}) = \frac{\pi_k^{(t)} N(X_i|\mu_k^{(t)}, \sigma_k^{(t)})}{\sum_{l=1}^{K} \pi_l^{(t)} N(X_i|\mu_l^{(t)}, \sigma_l^{(t)})}$$

Define:

$$\gamma_{i,k}(\Theta^{(t)}) = P(Z_i = k|X_i, \Theta^{(t)})$$

which represents the responsibility of component k for data point X_i.

Expectation Calculation (Q function) We compute the expectation:

$$Q(\Theta, \Theta^{(t)}) = \mathbb{E}_{Z|X,\Theta^{(t)}} [\log f(X, Z|\Theta)]$$

Expanding:

$$Q(\Theta, \Theta^{(t)}) = \sum_{i=1}^{n} \sum_{k=1}^{K} \gamma_{i,k}(\Theta^{(t)}) \left[\log \pi_k + \log N(X_i|\mu_k, \sigma_k^2)\right]$$

Remark: The inner part summation is because we have K Normal distribution components.

Substituting the normal density function:

$$Q(\Theta, \Theta^{(t)}) = \sum_{i=1}^{n} \sum_{k=1}^{K} \gamma_{i,k}(\Theta^{(t)}) \left[\log \pi_k - \frac{1}{2} \log 2\pi\sigma_k^2 - \frac{(X_i - \mu_k)^2}{2\sigma_k^2}\right]$$

1.4.3 M-Step: Parameter Updates

Updating Mean μ_k Taking the derivative with respect to μ_k:

$$\frac{\partial Q}{\partial \mu_k} = \sum_{i=1}^{n} \gamma_{i,k}(\Theta^{(t)}) \cdot \frac{X_i - \mu_k}{\sigma_k^2} = 0$$

Solving for μ_k:

$$\mu_k^{(t+1)} = \frac{\sum_{i=1}^{n} \gamma_{i,k}(\Theta^{(t)}) X_i}{\sum_{i=1}^{n} \gamma_{i,k}(\Theta^{(t)})}$$

Updating Variance σ_k^2 Taking the derivative with respect to σ_k^2:

$$\frac{\partial Q}{\partial \sigma_k^2} = \sum_{i=1}^{n} \gamma_{i,k}(\Theta^{(t)}) \left[-\frac{1}{2\sigma_k^2} + \frac{(X_i - \mu_k)^2}{2\sigma_k^4} \right] = 0$$

Solving for σ_k^2:

$$\sigma_k^{2(t+1)} = \frac{\sum_{i=1}^{n} \gamma_{i,k}(\Theta^{(t)})(X_i - \mu_k^{(t+1)})^2}{\sum_{i=1}^{n} \gamma_{i,k}(\Theta^{(t)})}$$

1.4.4 Summary

- **E-Step**: Compute posterior probability $P(Z_i = k|X_i, \Theta^{(t)})$, which represents the responsibility $\gamma_{i,k}(\Theta^{(t)})$.
- **M-Step**: Update parameters using weighted means:

$$\mu_k^{(t+1)} = \frac{\sum_{i=1}^{n} \gamma_{i,k}(\Theta^{(t)}) X_i}{\sum_{i=1}^{n} \gamma_{i,k}(\Theta^{(t)})}$$

$$\sigma_k^{2(t+1)} = \frac{\sum_{i=1}^{n} \gamma_{i,k}(\Theta^{(t)})(X_i - \mu_k^{(t+1)})^2}{\sum_{i=1}^{n} \gamma_{i,k}(\Theta^{(t)})}$$

- Iterate until parameter updates converge.
- The algorithm refines parameters iteratively to maximize the likelihood.

1.5 Why does it work?

1.5.1 Formula Expansion

We start from the fundamental decomposition:

$$\log f(X|\Theta) = \log f(X, Z|\Theta) - \log f(Z|X, \Theta)$$

which expresses the **marginal log-likelihood** in terms of the **complete data log-likelihood** and the **posterior distribution**.

Taking expectation over Z with respect to $P(Z|X, \Theta^{(t)})$:

$$\mathbb{E}_{Z|X,\Theta^{(t)}}[\log f(X, Z|\Theta)] - \mathbb{E}_{Z|X,\Theta^{(t)}}[\log f(Z|X, \Theta)]$$

Which expands to:

$$Q(\Theta, \Theta^{(t)}) + H(\Theta, \Theta^{(t)})$$

Where: - $Q(\Theta, \Theta^{(t)})$ is the **E-Step computed Q-function**. - $H(\Theta, \Theta^{(t)})$ is the entropy term:

$$H(\Theta, \Theta^{(t)}) = -\sum_{Z} P(Z|X, \Theta^{(t)}) \log P(Z|X, \Theta)$$

Thus, we obtain:

$$\log f(X|\Theta) = Q(\Theta, \Theta^{(t)}) + H(\Theta, \Theta^{(t)})$$

In the **EM procedure**, we set:

$$\Theta^{(t+1)} = \arg\max_{\Theta} Q(\Theta, \Theta^{(t)})$$

which means **the M-Step maximizes** $Q(\Theta, \Theta^{(t)})$ **to update parameters**.

1.5.2 Proving Log-Likelihood Increase

We want to show:

$$\log f(X|\Theta^{(t+1)}) - \log f(X|\Theta^{(t)}) \geq 0$$

From the previous step:

$$\log f(X|\Theta) - \log f(X|\Theta^{(t)}) = Q(\Theta, \Theta^{(t)}) - Q(\Theta^{(t)}, \Theta^{(t)}) + H(\Theta, \Theta^{(t)}) - H(\Theta^{(t)}, \Theta^{(t)})$$

Using **Jensen's inequality (or Gibbs inequality)**:

$$H(\Theta, \Theta^{(t)}) \geq H(\Theta^{(t)}, \Theta^{(t)})$$

We obtain:

$$\log f(X|\Theta) - \log f(X|\Theta^{(t)}) \geq Q(\Theta, \Theta^{(t)}) - Q(\Theta^{(t)}, \Theta^{(t)})$$

Remark: Gibbs inequality states that for all α and β such that $\sum_{i=1}^{n} \alpha_i = 1$, $\sum_{i=1}^{n} \beta_i = 1$, $0 \leq \alpha_i \leq 1$ and$0 \leq \beta_i \leq 1$, it holds that

$$\sum_{i=1}^{n} \alpha_i \log \beta_i \leq \sum_{i=1}^{n} \alpha_i \log \alpha_i$$

Since **the M-Step selects** $\Theta^{(t+1)}$ **to maximize** Q, we get:

$$Q(\Theta^{(t+1)}, \Theta^{(t)}) \geq Q(\Theta^{(t)}, \Theta^{(t)})$$

Thus, combining the results:

$$\log f(X|\Theta^{(t+1)}) - \log f(X|\Theta^{(t)}) \geq 0$$

This proves that **EM iterations monotonically increase the log-likelihood**, ensuring algorithm convergence.

So log-likelihood is guaranteed to increase within each step of the EM algorithm, unlike Newton-Raphson log-likelihood maximization.